

1 Server benchmark tools

Benchmark is using [k6](#).

```
/*
notes: 杨小兵-2024-12-19

1、对llama-server基准测试的tools
2、benchmark使用的是k6（目前对k6不熟悉）
*/
```

1.1 Install k6 and sse extension

SSE is not supported by default in k6, you have to build k6 with the [xk6-sse](#) extension.

```
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1、k6 默认不支持 SSE，你必须使用 [xk6-sse] 扩展来构建 k6。
2、使用xk6-sse来构建k6从而获取SSE的支持
*/
```

Example:

```
go install go.k6.io/xk6/cmd/xk6@latest
xk6 build master \
--with github.com/phymbert/xk6-sse
```

1.2 Download a dataset

This dataset was originally proposed in [vLLM benchmarks](#).

```
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1、下列命令行中的数据集最初是在[vLLM benchmarks]中被提出的
2、这个benchmark使用到了已有的数据集
*/
```

```
wget
https://huggingface.co/datasets/anon8231489123/ShareGPT_Vicuna_unfiltered/resolve/
main/ShareGPT_V3_unfiltered_cleaned_split.json
```

1.3 Download a model

Example for PHI-2

```
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1、这里下载PHI-2作为示例
2、下载模型文件格式是GGUF
*/
```

```
../../../../../scripts/hf.sh --repo ggml-org/models --file phi-2/ggml-model-q4_0.gguf
```

1.4 Start the server

The server must answer OAI Chat completion requests on <http://localhost:8080/v1> or according to the environment variable [SERVER_BENCH_URL](#).

```
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1、llama-server服务器必须回答`http://localhost:8080/v1` 上或根据环境变量
`SERVER_BENCH_URL`的OAI聊天完成请求
2、上述这段话是被翻译过来的，理解不到位
*/
```

Example:

```
server --host localhost --port 8080 \
  --model ggml-model-q4_0.gguf \
  --cont-batching \
  --metrics \
  --parallel 8 \
  --batch-size 512 \
  --ctx-size 4096 \
  -ngl 33
```

1.5 Run the benchmark

```
/*
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```

- 1、这部分内容介绍的内容就是运行benchmark
 - 2、具体的内容目前先不用关心
- */

For 500 chat completions request with 8 concurrent users during maximum 10 minutes, run:

```
./k6 run script.js --duration 10m --iterations 500 --vus 8
```

The benchmark values can be overridden with:

- `SERVER_BENCH_URL` server url prefix for chat completions, default `http://localhost:8080/v1`
- `SERVER_BENCH_N_PROMPTS` total prompts to randomly select in the benchmark, default `480`
- `SERVER_BENCH_MODEL_ALIAS` model alias to pass in the completion request, default `my-model`
- `SERVER_BENCH_MAX_TOKENS` max tokens to predict, default: `512`
- `SERVER_BENCH_DATASET` path to the benchmark dataset file
- `SERVER_BENCH_MAX_PROMPT_TOKENS` maximum prompt tokens to filter out in the dataset: default `1024`
- `SERVER_BENCH_MAX_CONTEXT` maximum context size of the completions request to filter out in the dataset: prompt + predicted tokens, default `2048`

Note: the local tokenizer is just a string space split, real number of tokens will differ.

Or with [k6 options](#):

```
SERVER_BENCH_N_PROMPTS=500 k6 run script.js --duration 10m --iterations 500 --vus 8
```

To [debug http request](#) use `--http-debug="full"`.

1.6 Metrics

Following metrics are available computed from the OAI chat completions response `usage`:

- `llamacpp_tokens_second` Trend of `usage.total_tokens` / request duration
- `llamacpp_prompt_tokens` Trend of `usage.prompt_tokens`
- `llamacpp_prompt_tokens_total_counter` Counter of `usage.prompt_tokens`
- `llamacpp_completion_tokens` Trend of `usage.completion_tokens`
- `llamacpp_completion_tokens_total_counter` Counter of `usage.completion_tokens`
- `llamacpp_completions_truncated_rate` Rate of completions truncated, i.e. if `finish_reason === 'length'`
- `llamacpp_completions_stop_rate` Rate of completions stopped by the model, i.e. if `finish_reason === 'stop'`

The script will fail if too many completions are truncated, see `llamacpp_completions_truncated_rate`.

K6 metrics might be compared against [server metrics](#), with:

```
curl http://localhost:8080/metrics
```

2 Using the CI python script

The `bench.py` script does several steps:

- start the server
- define good variable for k6
- run k6 script
- extract metrics from prometheus

It aims to be used in the CI, but you can run it manually:

```
LLAMA_SERVER_BIN_PATH=../../cmake-build-release/bin/llama-server python  
bench.py \  
    --runner-label local \  
    --name local \  
    --branch `git rev-parse --abbrev-ref HEAD` \  
    --commit `git rev-parse HEAD` \  
    --scenario script.js \  
    --duration 5m \  
    --hf-repo ggml-org/models \  
    --hf-file phi-2/ggml-model-q4_0.gguf \  
    --model-path-prefix models \  
    --parallel 4 \  
    -ngl 33 \  
    --batch-size 2048 \  
    --ubatch-size 256 \  
    --ctx-size 4096 \  
    --n-prompts 200 \  
    --max-prompt-tokens 256 \  
    --max-tokens 256
```